Homework: Attention Mechanism and Transformer

Q1. Discuss two major limitations of conventional RNNs when processing long sequences. Explain how the attention mechanism helps address the limitation. An RNN encoder produces the following hidden states for an input sequence of three tokens:

$$h_1 = [1,2], \quad h_2 = [2,1], \quad h_3 = [3,2]$$

For a particular decoder time step, the attention weights are:

$$\alpha_{t,1} = 0.2, \quad \alpha_{t,2} = 0.5, \quad \alpha_{t,3} = 0.3$$

Calculate:

a) Verify that the attention weights satisfy the normalization property.
b) Calculate the weighted contribution of each encoder hidden state.
c) Calculate the context vector c_t.
d) Identify which encoder hidden state contributes most to the context vector.

Q2. Define self-attention and explain how self-attention captures long-range contextual relationships between words. Explain the roles of Query (Q), Key (K), and Value (V) vectors in the self-attention mechanism. Derive the scaled dot-product attention equation and explain the purpose of the scaling factor.

Consider a query vector and two key vectors: $Q = [1,0], \quad K_1 = [1,0], \qquad K_2 = [0,1]$

The corresponding value vectors are: $V_1 = [2,1], \qquad V_2 = [0,3]$

Using scaled dot-product attention, calculate:

a) The dot product between the query and each key vector.
b) The scaled attention scores using d_k=2.
c) Apply softmax to calculate the attention weights.
d) Calculate the final attention output as the weighted sum of the value vectors.

Q3. Explain the major components of the Transformer architecture with a suitable block diagram. Explain the functions of self-attention, cross-attention, and feed-forward networks in a Transformer. Consider a Transformer model has a hidden dimension of 512 and uses 8 attention heads. Assume that the input sequence contains 10 tokens.

Calculate:

a) The dimension of Query, Key, and Value vectors for each attention head.
b) The dimensions of the Query, Key, and Value matrices for one attention head.
c) The dimensions of the attention score matrix for one head.
d) Calculate the total number of attention score values generated across all 8 heads.
e) Explain why multi-head attention is more effective than a single attention head.

Q4. Why do Transformers require positional encoding, whereas RNNs inherently process tokens sequentially. Explain how positional encoding is combined with token embeddings in transformer. Consider a token embedding in a transformer model:

$$X = [0.2, 0.5, 0.3, 0.1]$$

and its positional encoding:

$$P = [0.1, 0.2, 0.4, 0.3]$$

Calculate the resulting representation after adding positional encoding to the token embedding. Differentiate between fixed and learned positional encodings.

Q5. Explain how sparse attention reduces the number of token-to-token attention connections compared with standard self-attention. Consider A Transformer is required to process long sequences while maintaining computational efficiency and capturing important contextual relationships.

a) Explain why standard dense attention is inefficient for long sequences.
b) Design a sparse attention pattern using local and global attention.
c) Explain how your proposed design captures both nearby and distant contextual information.
d) Discuss one limitation of the proposed design.

Q6. What is masked self-attention? Explain why future tokens must be masked during autoregressive text generation. Consider the sequence: “The student is reading a book.”

a) Explain which tokens are accessible to the model when predicting the next token after "reading" under causal masking.
b) c) Explain how an autoregressive Transformer generates a sequence one token at a time.
c) Differentiate between masked self-attention in GPT and bidirectional self-attention in BERT.

Q7. Compare the architectures of BERT and GPT in terms of encoder and decoder components. Explain the training objectives of BERT and GPT. Why is BERT generally suitable for language understanding tasks, whereas GPT is suitable for autoregressive text generation? Give two application examples each for BERT and GPT.

Q8. Explain the basic architecture of Vision Transformer (ViT). How are image patches converted into tokens for processing by a Transformer? Consider an RGB image of size 224×224 pixels, divided into non-overlapping patches of size 16×16. Calculate the total number of image patches. Explain the role of positional information and the encoder in ViT-based image classification.

Q9. A Transformer processes the sentence: "Deep learning improves image recognition."

Assume:

- Number of tokens = 5
- Embedding dimension = 128

Calculate:

a) The dimension of the embedding vector for each token.
b) The dimension of the complete input embedding matrix.
c) If positional encoding of dimension 128 is added to each token embedding, determine the dimension of the resulting matrix.
d) If the Transformer uses 4 attention heads, calculate the dimension of each attention head.
e) Explain why the positional encoding and token embedding must have compatible dimensions for element-wise addition.

Q10. Consider a Transformer encoder-decoder model used for machine translation.

- Source sequence length = 5 tokens.
- Target sequence length = 3 tokens.
- Hidden dimension = 512.
- Number of attention heads = 8.

Calculate:

a) The dimension of each attention head.
b) The dimensions of the Query, Key, and Value matrices in encoder self-attention.
c) The dimensions of the Query, Key, and Value matrices in decoder cross-attention.
d) Calculate the dimensions of the attention score matrices for encoder self-attention and decoder cross-attention, respectively.
e) Explain why the Query matrix in cross-attention comes from the decoder, whereas Key and Value matrices come from the encoder